Held-out prediction agreement by marker

Per-slide P99 normalization; stride-32 training; expression-weighted MLP

Pearson correlation

PODXL	0.718
ZO1	0.608
CD10	0.579
VIM	0.577
PDPN	0.569
CAV	0.561
PCNA	0.550
NES	0.520
PanCK	0.516
CDH1	0.511
CD34	0.505
UMOD	0.491
CD31	0.489
CD138	0.458
CTNNB1	0.455
BCL2	0.450
IFNA6	0.449
CD4	0.430
aSMA	0.416
DAPI	0.406
CDK1	0.386
PTEN	0.360
STAT1	0.318
AQP2	0.314
CD93	0.314
HIF1a	0.287
GATA3	0.286
ATM	0.267
PARP1	0.246
Cd11c	0.240
CD8	0.233
COLIV	0.230
CD45	0.214
Ki67	0.214
NF2	0.208
HIF2a	0.207
CAIX	0.204
POSTN	0.179
CD14	0.171
PAX2	0.159
CD1a	0.133
IL6	0.101

0.0 0.2 0.4 0.6 0.8 1.0

Patient-mean correlation

Spearman rank correlation

PODXL	0.202
ZO1	0.278
CD10	0.594
VIM	0.602
PDPN	0.196
CAV	0.487
PCNA	0.573
NES	0.325
PanCK	0.286
CDH1	0.383
CD34	0.460
UMOD	0.265
CD31	0.444
CD138	0.608
CTNNB1	0.472
BCL2	0.458
IFNA6	0.433
CD4	0.342
aSMA	0.190
DAPI	0.475
CDK1	0.396
PTEN	0.331
STAT1	0.286
AQP2	0.183
CD93	0.304
HIF1a	0.293
GATA3	0.189
ATM	0.236
PARP1	0.340
Cd11c	0.229
CD8	0.277
COLIV	0.285
CD45	0.131
Ki67	0.204
NF2	0.231
HIF2a	0.286
CAIX	0.174
POSTN	0.088
CD14	0.137
PAX2	0.155
CD1a	0.126
IL6	0.114

0.0 0.2 0.4 0.6 0.8 1.0

Patient-mean correlation

42 markers; 39 patients. Equal-patient means of within-patient OOF metrics.
Bars: reported 95% patient-bootstrap CIs, conditional on trained predictions; shared-fold training uncertainty is not fully captured.